1. Azure RBAC (Role Based Access Control)

Purpose:

Controls **who can access Azure resources and what actions they can perform**.

Scope:

Works on **Azure resources** like:

- Virtual Machines
- Storage Accounts
- Resource Groups
- Subscriptions

Example Actions Controlled

- Create VM
- Delete VM
- Read Storage
- Modify Network

How it works

You assign a **role** to a **user, group, or service principal**.

Example roles:

- Owner
- Contributor
- Reader
- Virtual Machine Contributor
- Network Contributor

RBAC Scope Hierarchy

```
Management Group
    ↓
Subscription
    ↓
Resource Group
    ↓
Resource
```

Example

User Priyansh → Contributor → Resource Group

Result:

- Can create VM
- Can modify VM

Permissions **inherit downward**.

2. Azure Policy

Purpose:

Controls **what is allowed or not allowed to be created in Azure**.

It enforces **rules and compliance**.

Scope:

Applies to **resources and configurations**.

Example Rules

- Only allow **specific VM sizes**
- Only allow **East US region**
- Require **tags**
- Enforce **encryption**
- Block public IP creation

Policy Effects

Common effects:

- **Deny** → Block resource creation
- **Audit** → Allow but report non-compliance
- **Append** → Add settings automatically
- **Modify** → Modify configuration
- **DeployIfNotExists** → Deploy missing resources

Example

Policy:

```
Allow VM only in West Europe
```

Result:

- User tries creating VM in East US → ❌ Denied

Even if user is **Owner**, policy still blocks it.

Key point

RBAC gives **permission**,

Policy controls **rules**.

3. Entra ID Roles (Azure AD Roles)

Purpose:

Controls **identity and directory management permissions**.

This is **NOT for Azure resources**.

These roles manage:

- Users
- Groups
- Authentication
- Security settings
- Identity management

Examples:

- Global Administrator
- User Administrator
- Security Administrator
- Application Administrator
- Billing Administrator

Example

User Administrator can:

- Create users
- Reset passwords
- Manage groups

But cannot create Azure VMs.

Feature	RBAC	Azure Policy	Entra ID Role
Purpose	Access control	Governance & compliance	Identity management
Controls	Actions on resources	Resource configuration	Directory management
Example	Create VM	Restrict VM size	Create users
Applied To	Azure resources	Azure resources	Users & identity
Level	Resource hierarchy	Resource hierarchy	Tenant level

Think like this:

RBAC → WHO can do WHAT

Policy → WHAT is allowed

Entra Role → WHO can manage USERS